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1- All the following are tools needed to genetically manipulate microorganisms for production desired products except: Select one:

- a. None of these
- b. Promoter that controls the expression of the gene
- c. Markers for selection of the recombinant protein**
- d. Cloning vectors
- e. Methods for gene transfer

2- All the following methods are used for gene transfer except: Select one :

- a. Transduction
- b. Electroporation
- c. None of these**
- d. Transformation
- e. Protoplast transformation

3- Solid substrate fermentation involves the following except : Select one or more:

- a. None of these
- b. Growing bacteria on these hydrolysis products**
- c . Mechanical or chemical processing of the solid media
- d. Hydrolysis of polysaccharides and proteins
- e. Purification of the end product

4- The following are methods for sludges disposal except; Select one:

- a. Landfilling
- b. Incineration
- C. Thickening
- d. Dehydration
- e. Trickling**

**5-All the following are processes of the product purification (down streaming) except:
Select one:**

- a. Dialysis
- b. Partition
- c. Electrophoresis**
- d. Distillation
- e. Ultra filtration

6- All the following are disadvantages of the continuous fermentation except : Select one :

- a. Is not acceptable by US FDA especially in the production of therapeutic products
- b. All products are produced optimally which might lead to formation of inclusion bodies**
- c. Mutations might easily occur while bacteria is growing for long time
- d. None of these
- e. Contamination can be a major problem

7- The good designed pesticides should fulfill all the following except: Select one:

- a. Can be sprayed evenly on the plants
- b. Effective at very low concentrations
- c. None of the choices is wrong**
- d. Does not accumulate and ruin the environment
- e. Not be harmful to the end user

8- All the following are applied for genetically modified microorganisms EXCEPT

Select one:

- a. Amenability to genetic modification
- b. Ready breakage if target is intracellular
- c. Production of limited non wanted products**
- d. Produce no toxic material at all
- e. Able to grow on low cost media

9- Oxygen transfer is influenced by all the following factors except : Select one :

- a. Material the sparger is made of
- b. Temperature
- c. Speed of the agitation
- d. Chemical nature of the media
- e. Surface area of air/oxygen bubbles

10- All the following applies to the stationary phase except: Select one :

- a. No growth
- b. All of these
- c. Accumulation of waste, pH changes , and exhaustion of the nutrients

d. Production of primary metabolites

- e. A balance in the number of multiplying and dying

11- One of the following statements is wrong; Select one:

a. Zootoxins are toxic only to zoo animals

- b. Allelochemicals are organic compounds produce by microbes or plants to work against plant microbes
- c. Phytotoxins are toxic to plants
- d. Antibiotics can be used against plant pathogens
- e. Allelopathy are biochemical interactions occurring among plants

12- Which of the following is not a heavy metal ? Select one :

- a. Mercury
- b . Lead
- c. Magnesium**
- d . Chromium
- e . Cadmium

13- All the following conditions should be present in a microbial pesticide except;

- a. can easily be distributed using water
- b. does not accumulate in the environment
- c. should not be harmful to the first user
- d. can be used on plants without difficulty
- e. can be effective in lowest possible concentration

14. in making Sauerkraut and pickling salt is added to :

- a. to reduce water activity which suppressed the growth of gram negative organisms
- b. only to reduce water activity so that gram positive such as Staphylococcus spp. cannot grow
- c. to inhibit the growth of both Gram negative and Gram Positive bacteria
- d. just to give the salty taste to the pickles
- e. promote the growth of Lactic acid bacteria that helps in pickling

15- One of the following statements is not correct about cheese making :

- a. removing the k caseins exposed the non- polar surface of the micelle which helps in the curd formation
- b. upon the addition of chymosine, the amino acids 105-106 are broken and k casein is separated
- c. the presence of calcium is vital for the none polar micelles to bind together
- d. none of these
- e. rennin removes surface glycopeptides from the k casein thus promoting the collapse of the micelle

16- One of the following statements is not correct ;

- a. aerobic bacteria are fast growing, minutes are needed while hours are needed for the anaerobic bacteria
- b. only small amount of carbon is converted into cell mass
- c. None of these
- d. in aerobic wastewater treatment about 50% of biodegradable carbon is converted in biomass while the other half if converted into CO₂
- e. in the aerobic growth , no energy is produced while energy is produced in the anaerobic

17- The following techniques of the in situ bioremediation except ;

- a. attenuation
- b. biopileing
- c. phytoremediation
- d. bioaugmentation**
- e. biostimulation

18- Which of the following is not nitrogen fixing microorganism;

- a. Rhizobium
- b. Frankia
- c. Cyanobacteria
- d. Shewanella**
- e. Azotobacter

19- All the following are microbial approaches except:

- a. microbial consortia
- b. cultivation
- c. metabolomics**
- d. biodiversity
- e. microbe -microbe interaction

20- All the following are steps in the cheese making process except :

- a. ripening and ageing
- b. sterilizing the milk prior to cheese making so that no pathogenic**
- c. removal of the whey
- d. formation of the curd
- e. none of these

21- During the ingestion of the toxin and spores, the insect will die due to :

- a. all of these
- b. the action of both toxin and spore germination
- c. toxin activation inside the gut of the worm
- d. none of these
- e. the spore germination inside the bacteria

22- All the following microorganisms can be used as a starter culture except ;

- a. *Leuconostoc mesenteroides*
- b. *Lactobacillus subtilis*
- c. *Streptococcus thermophilus*
- d. *Lactobacillus helveticus*
- e. *Lactococcus lactis*

23- When producing penicillins, the acylase treatment is used to;

- a. purify the product as this enzyme helps in removing contaminants
- b. to extract the penicillin from the cell free material
- c. to produce the semisynthetic penicillin
- d. purify the crystals from any crystallized contaminant.
- e. crystalize the product

24- All the following are strategies to modify the terminal pathway EXCEPT :

- a. introduction of more of the native enzyme to increase
- b. amplification of rate limiting enzyme
- c. amplification of the branch-point enzyme
- d. none of these
- e. amplification of the first enzyme in the pathway

25- All these are mechanisms of the biofungicides except :

- a. parasitism
- b.direct competition
- c. inducing fungi resistance to the host plant
- d . antagonism

e.none of these

26- The term secondary metabolites means they are produced in the following growth phase;

- a . the extended death phase

b. stationary phase

- c. log phase
- d. decline phase
- e. lag phase

27- All the following are applications of microorganisms in the environment except;

- a. none of these

b. biodegradation of xenobiotic (inorganic compounds)

- c . biomining
- d. bioremediation of heavy metals
- e. waste water treatment

28- Which among the following statements is incorrect ;

- a. Tryptophan is used as antidepressant
- b. Arginine is used for ammonia detoxification
- c. Methionine deficiency lead to fatty liver and liver cirrhosis

d. Ornithine used in the treatment of neuropathic diseases

- e. Histidine helps in hemoglobin regeneration

29- In making cheese from milk, the caseins forms the major part of the cheese while the whey proteins are used in small amounts.

- a . True

b . False

30- Which among the following are most limiting nutrients for plant growth and development;

- a. None of these
- b. Sulphate for building amino acids
- c. Ammonia and phosphate
- d. ATP and ADP
- e . Cupper and zinc

31- The agent Acarbose inhibit enzyme a glucosidase that prevents the absorption of oligosaccharides and polysaccharides in the intestine.

- a. True
- b. False

32- One of the following statements is not correct;

- a. Hydrolysis of complex organic compounds occurs by extracellular fungal enzymes
- b. Methanogenesis production of methane from acetic acid, H₂ and CO₂
- c. none of these
- d. acidogenesis occurs due to transformation of sugars, fatty acids and amino acids
- e. Acetogenesis occurs due to transformation of higher organic acids

33- All the following are specific uses of microorganisms in the environment except;

- a. improving soil aggregation by solubilizing the available polysaccharides
- b. suppression of soil pathogens
- c. production of bioactive compounds
- d. decomposing dead plants and animals
- e. nitrogen fixation

34- One of the following statements is not correct regarding food fermentation;

- a. Koji is fermented by *Aspergillus oryzae*
- b. Moromi is fermented by *Pediococcus halophilus*
- c. *Aspergillus oryzae* is used to hydrolyze starch, proteins and pectin
- d. Tempeh is fermented by bacteria belongs to *Rhizopus* spp.
- e. furanones are flavor enhancing compounds on the soy bean upon fermentation

35- All the following are health products produced by microbes except;

- a. anti-viral agents such as Cyclovir
- b. anti-lipidemic agents such as Mevastatin
- c. anti-diabetic agents such as Valiolamine
- d. anti-tumor agents
- e. agents that suppress the immune system by interfering with activity and growth of T cells

36- Which among the following is used as a sweetener for diabetic patients

- a . none of these
- b. cobalamin anhydrous
- c. Monosodium glutamate
- d . amino acid tryptophan
- e. Splenda

37- Which of the following is not an environmental application of microbes?

- a. Biofertilizers
- b. Biodegradation of xenobiotic compounds
- c. Waste water treatment
- d. Bioremediation
- e. Bioremediation of heavy metals

38- The following are bacterial antibiotic targets except :

- a. cell wall
- b. efflux pathway
- c. ribosomes
- d. folate synthesis pathway
- e. DNA replication

39- All the following are processes utilized in the finishing product (down streaming) EXCEPT :

- a. crystallization
- b. centrifugation**
- c. sterile filtration
- d. Lyophilization
- e. Evaporation

40- Tocopherols are antioxidants mainly produced by : Select one :

- a. None of these
- b . Cyanobacterium**
- c . Fungi
- d. Archaea
- e. Photosynthetic bacteria

41- When comparing biostimulation to bioaugmentation the biostimulation is superior as fresh nutrients are all what microbes need Select one :

- a. True
- b. False**

42- the most important protein involved in the acid coagulation of milk is: Select one :

- a. kappa caseins**
- b. gamma caseins
- c. beta caseins
- d. alpha caseins
- e. all of these

43- Antibiotics are :

Secondary metabolite produced by bacteria in the stationary phase

44- one of the following statements is correct regarding the Biochemical Oxygen (BOD):

Select one:

- a. the amount of dissolved oxygen needed by anerobic bacteria to breakdown organic matter present in a given water sample
- b. the amount of atmospheric oxygen that is needed for certain biochemical reactions
- c. BOD is a gauge for measuring the effectiveness of bacteria to degrade inorganic materials in wastewater tanks
- d. the amount of dissolved oxygen needed by aerobic bacteria to breakdown organic matter in a given water sample
- e. high values means lower pollution in the sample

45- The amino acid ornithine is used for the following: Select one :

- a. Helping in the treatment of duodenal ulcer
- b. Protect the SH enzymes in the liver from inhibit
- c. Used as an anti -depressant
- d. Ammonia detoxication
- e. Sweetener for diabetics

46- The following are whey proteins except : Select one :

- a. blood serum proteins
- b. beta lactoglobulin
- c. alpha lactoglobulin
- d. immunoglobulins
- e. gama lactoglobulin

47- Lactic acid bacteria have the following effects except : Select one:

- a. lead to shrinkage of the whey protein and thus lead to more cheese yield
- b. decrease the pH which kills some of the pathogenic bacteria
- c. produces ketones, esters and aldehydes as flavoring agents
- d. none of these**
- e. produce lactic acid which makes rennin work better

48- The vitamin that is produced only by bacteria is Select one:

- a. Thiamine
- b. Vitamin
- c. Cobalamin**
- d. Beta carotene
- e. Vitamin A

49- One of the following matches regarding fermentation is not correct : Select one:

- a. None of these
- b. Koji is fermented by *Aspergillus oryzae*
- c. Coffee beans is fermented by filamentous fungi
- d. *Candida albicans* produce alcoholic fermentation to moromi thus produce flavoring compounds**
- e. Moromi is fermented by *Pediococcus halophilus*

50- Biosorption is a process that mean. e Select one :

- a. Solubilizing the metals and thus making them easy to remove
- b. All of these process are considered biosorption
- c. Microorganisms metabolize metals and remove from environment
- d. Is an active process where microorganisms spend high amount of energy to finish the sorption
- e. binding of metal ions to bacteria and eliminating them**

51- the following could be included in the group of endophytes except: Select one:

a. Actinomycetes

b. Bacteria

c. Fungi

d. Streptomyces

e. all of these

52- Flocs contain all the following except: Select one:

a. None of these

b. Organic matter .

c. Extracellular polymeric substances

d. Inorganic matter

e. Bacterial cells

53- Incorporating microbial genes into crop plants confer all the following except : Select one :

a. Ability to grow in harsh environment

b. Resistant to insect pests

c. Herbicide tolerance

d. Control of pathogenic microbes

e. None of these

54- Fungi can be controlled by following mechanisms EXCEPT :

a. Parasitism

b. inducing resistance of the fungi to host plant

c. Direct competition

d. None of these

e. Antagonism

55- All the following are xenobiotic compounds except ; Select on :

- a. None of these
- b. Nitro-aromatic compounds
- c. Hydrocarbons in general
- d. Phthalate esters
- e. Halogenated alephatic compounds

56- All the following bacteria are used as bioinsecticides except Select one:

- a. Bacillus anthrax
- b. Bacillus subtilis
- c. Bacillus lenimorbus
- d. Bacillus thuringiensis
- e. Bacillus sphaericus

57- One of the following is not considered as fermentation: Select one:

- a. Production of the wine
- b. Mass culture of anaerobic microorganisms
- c. Biological processes that occur in the absence of Oxygen
- d. Purification of foods especially proteins
- e. All of these conform to the definition of the fermentation

58- All the following bacterial species that are involved in the production of amino acids except :

- a. Brevibacterium spp
- b. Corynebacterium spp.
- c. Mycobacterium spp.
- d. Microbacterium spp.
- e. Arthrobacter spp

59- All the following parameters remain constant during the continuous fermentation using the chemostat with a changing dilution rates except ;

- a. pH of the growing medium
- b. temperature
- c. volume of the culture
- d. biomass of the microorganisms

e. none of these

60- In 1922 the first genetically engineered corn was produced and sold commercially.

Answer: **True**

61- Protoplast fusion is a method used to introduce the naked DNA into cells.

Answer: **False**

62- Secondary metabolites might be produced in the fed-batch fermenter but not the continuous fermenter.

Answer: **False**

63-All the following processes are considered downstream processing except:

Answer: **strain improvement**

64- All the following are considered concentration steps in the downstream processing except:

Answer: **depth filtration**

65- The incremental feeding regimes in the fed- batch process is considered the best regime of feeding

Answer: **True**

66- All the following are disadvantages of batch fermentation system except:

Answer: **acceptable by FDA when used for drug or food production**

67- Who developed the site directed mutagenesis:

Answer: **Michael Smith**

68- Which of these sentences is not correct:

Answer: **allelochemicals are inorganic compounds produced by microorganisms against other organisms**

